LIÈGE université
Sciences Appliquées

# Optimization module overview

## ELEN0445 - Microgrids

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# You Have Learned

- What is a microgrid (structure)
- What are its main components
- Power electronics interfaces (MPPT, etc.)
- The levels of control
- How to design an installation using SMA's software

# Optimal Control and Sizing

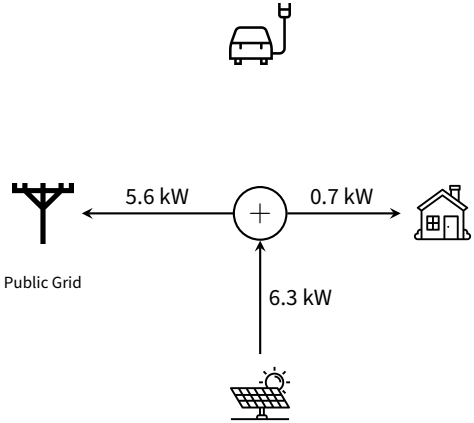
**Objective:** Learn how to optimally size and control a microgrid.

We will still focus on a one-node microgrid for illustration

Possible problem extensions:

- **Microgrid networks:** Adding power flow equations (spatial complexity).
- **Multi-energy systems:** District heating, Hydrogen ($H_2$), etc..
- **Community microgrids:** Fairness and sharing considerations.

# Real-time Control:

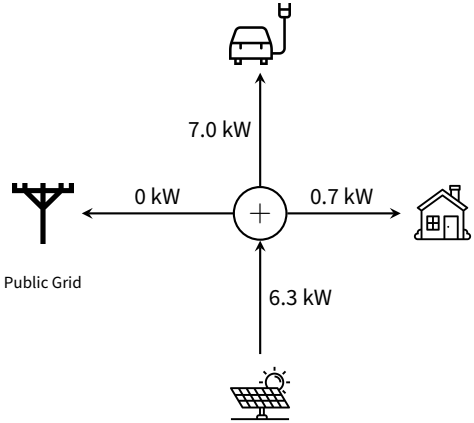



Max grid exchange: **11.0 kW**

**Sequence of events:**

1. Initial: Surplus exported to grid.

# Real-time Control:




Max grid exchange: **11.0 kW**

**Sequence of events:**

1. Initial: Surplus exported to grid.
2. **Car connects**: Adding 7 kW load.

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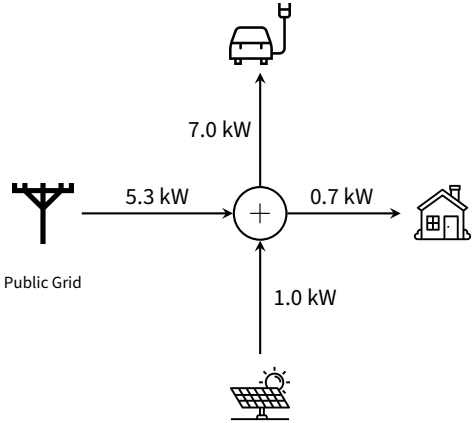


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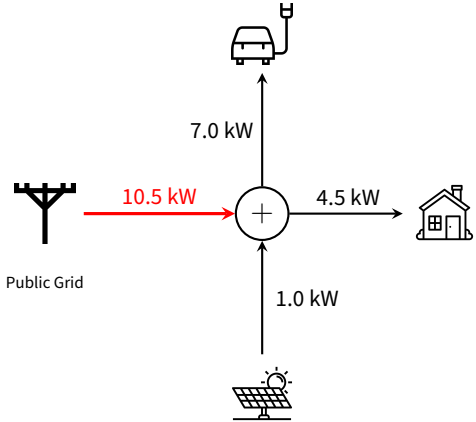



Max grid exchange: **11.0 kW**

**Sequence of events:**

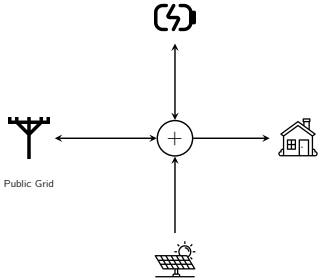
1. Initial: Surplus exported to grid.
2. **Car connects**: Adding 7 kW load.
3. **Cloud passes**: PV drops to 1 kW.
4. **Peak load**: House rises to 4.5 kW.

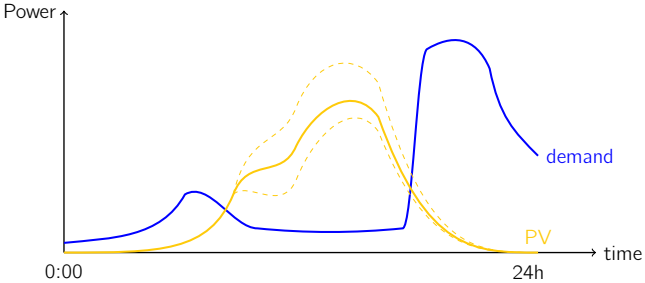
**Decision Point**

Import is at 10.5 kW. Any small increase will trip the 11 kW limit.

**Should you stop charging?**

# Operational Planning (E.g. 24h ahead)




- Based on **forecasted demand and PV generation**.
- **Decisions:** Store, send back to grid, or move load?
- **Variables:** Function of prices, physical limits, and efficiencies.

# Sizing (Investment Decisions)

Strategic investment questions:

- Should you connect to the grid? Which power?
- Should you buy a genset or a CHP (Combined Heat and Power)?
- Should you invest in PV or wind generation?
- Should you invest in storage?

**Interdependence**

Sizing decisions depend directly on the operational planning policy.

# Coordination of Time Scales

The global problem is decoupled in practice:

| **Horizon** | **Decision Type** |
|---|---|
| Yearly | Investment (Sizing) |
| Hours/Days | Planning (Operational Planning) |
| Minutes/Seconds | Adjustment (Real-time Control) |

# Example: coordination between real-time control and operational planning

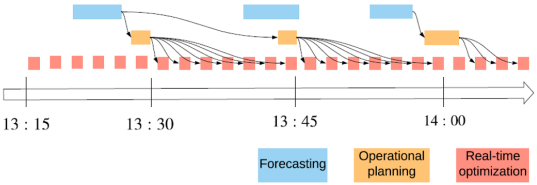



Fig. 1: Hierarchical control procedure illustration.

References:


- Dumas, J., Dakir, S., Liu, C., & Cornélusse, B. (2021). Coordination of operational planning and real-time optimization in microgrids. Electric Power Systems Research, 190, 106634.
- Moureau, C., Stegen, T., Glavic, M., & Cornélusse, B. (2025). A Predictive Flexibility Aggregation Method for Low Voltage Distribution System Control. ORBi-University of Liège.

# Goals of this module

- Learn about mathematical programming
- Learn to model problems as LPs or MIPs
- Understand how solvers work
- Solve the real-time control problem
- Learn to make some forecasts
- Solve the operational planning problem
- Solve the sizing problem